Report on Practical 5

Static Routing

1. Aims

- Implement a network using Cisco Packet Tracer, configure static routing to enable data transfer between devices in different router.

2. Objectives

- Understand the importance of static routing in small-scale networks.
- Assign appropriate IP addresses to devices for effective data transfer.
- Configure static routes on routers to transfer data between different networks.
- Verify routing configurations by testing connectivity between PCs.

3. Background

- **Routing** involves selecting paths in a network along which data packets travel and ensures that packets are efficiently delivered across networks
- Routing has two primary types of routing method, which are:

    Static routing

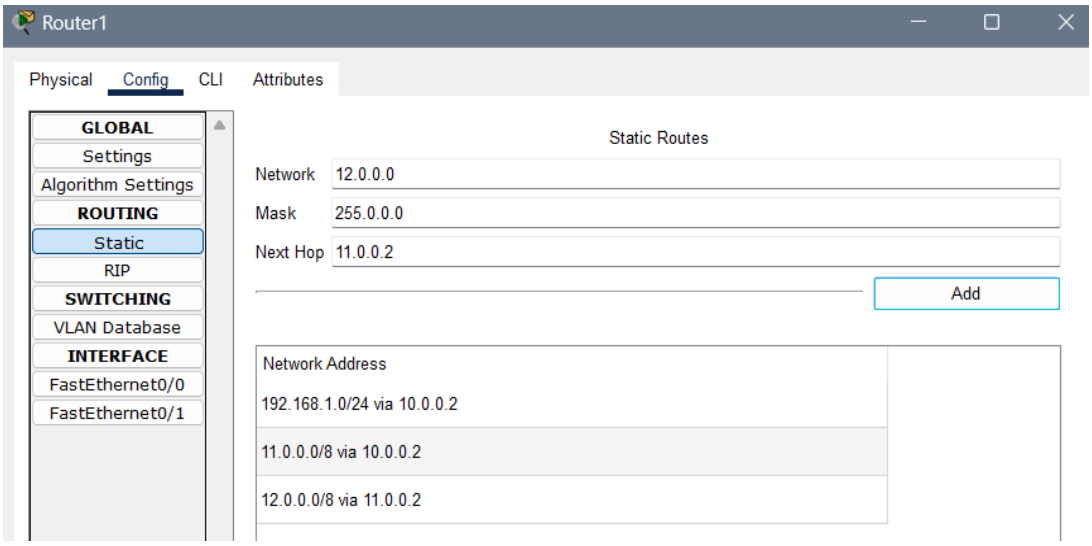
    Dynamic routing

- Dynamic routing is a method where routers automatically exchange information using routing protocols to learn and adapt to changes in the network.
- It is mostly used larger, more complex, or frequently changing networks.
- Static routing is an IP-based routing system that relies on manually configured routing tables to function.
- Unlike dynamic routing protocols, static routing provides more control and is useful for smaller, simpler networks.
- It is also more secure as routes are not advertised across the network and no overhead from routing protocol updates.

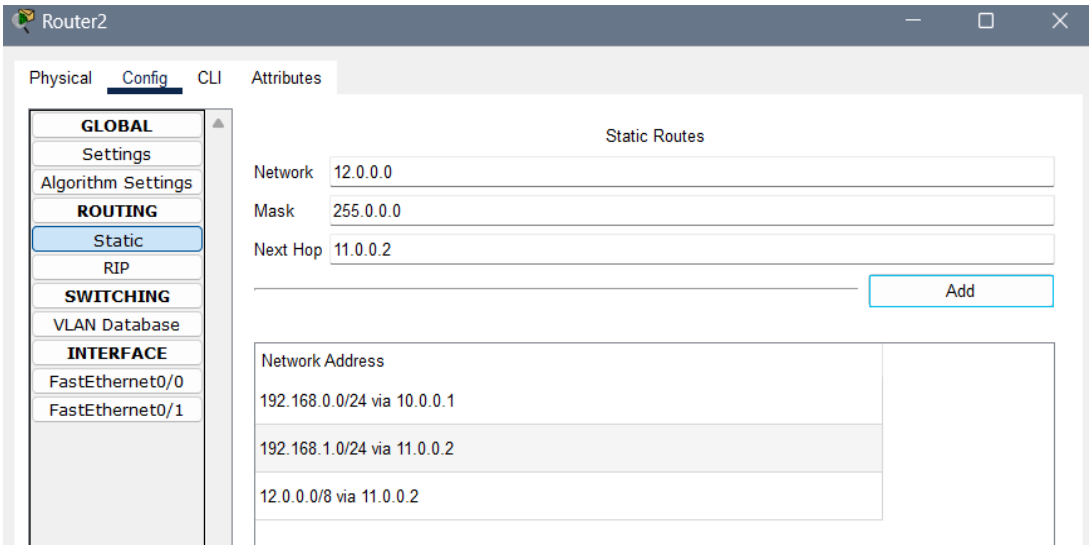
4. Procedure

1. Access the conjuration settings for each router in the network.

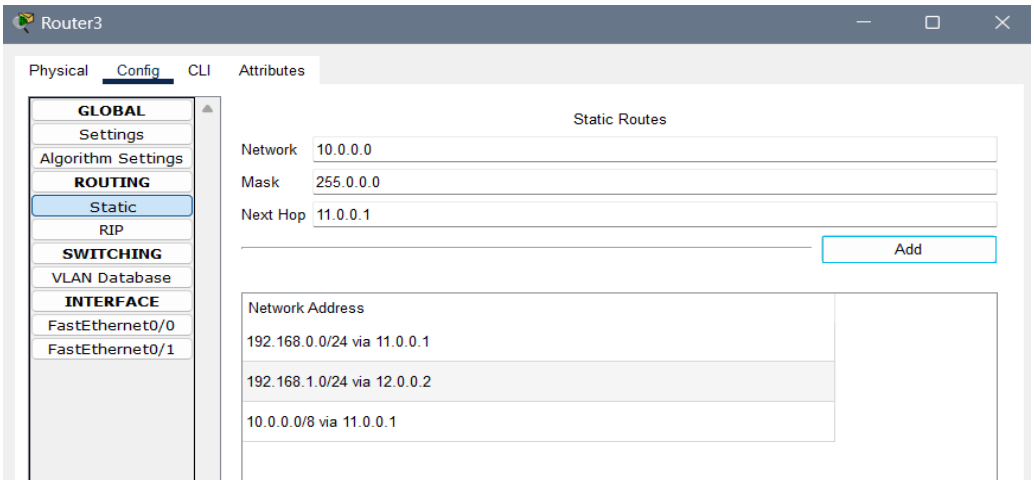
2. On Router 1, configuration static routes to reach the networks connected to Routers 3 and 4. Specify the appropriate next-hop IP addresses.

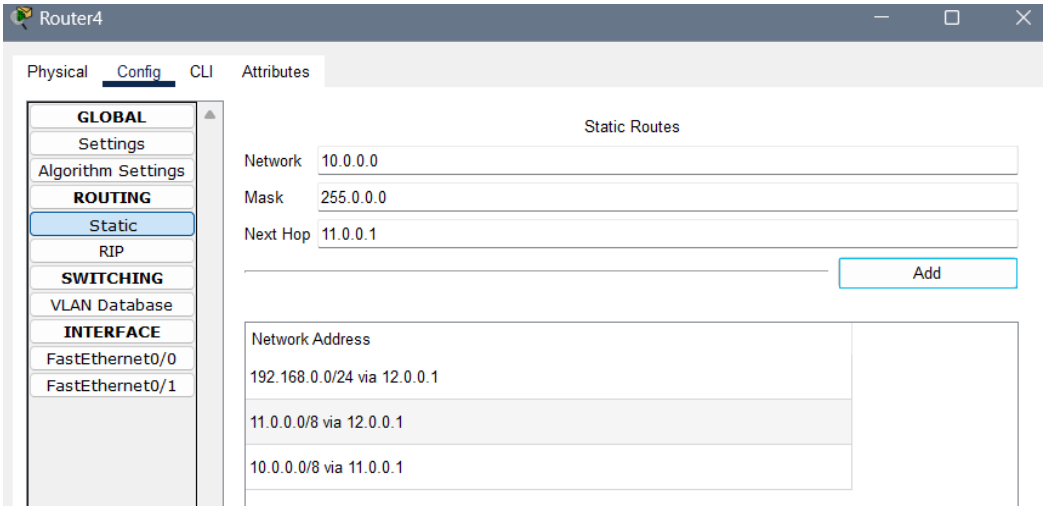


3. On Router 2, configure a static route to reach the network connected to Router 1 using the correct next-hop IP address.

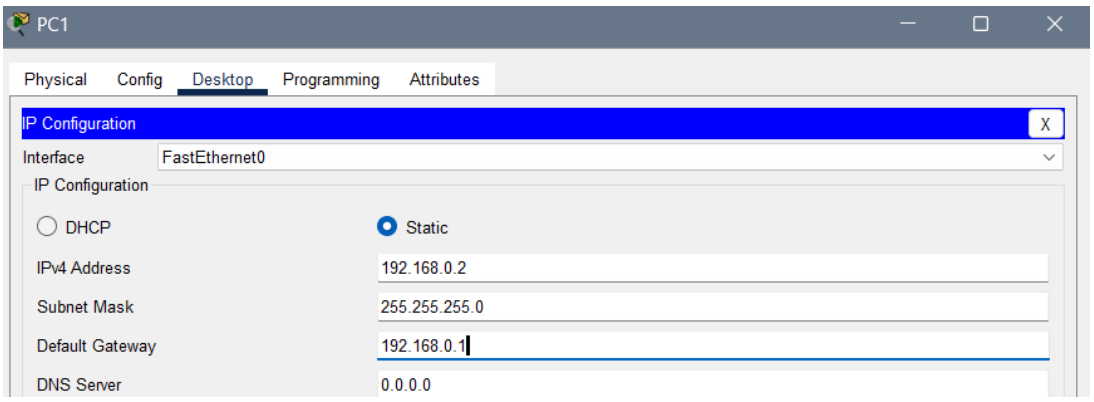


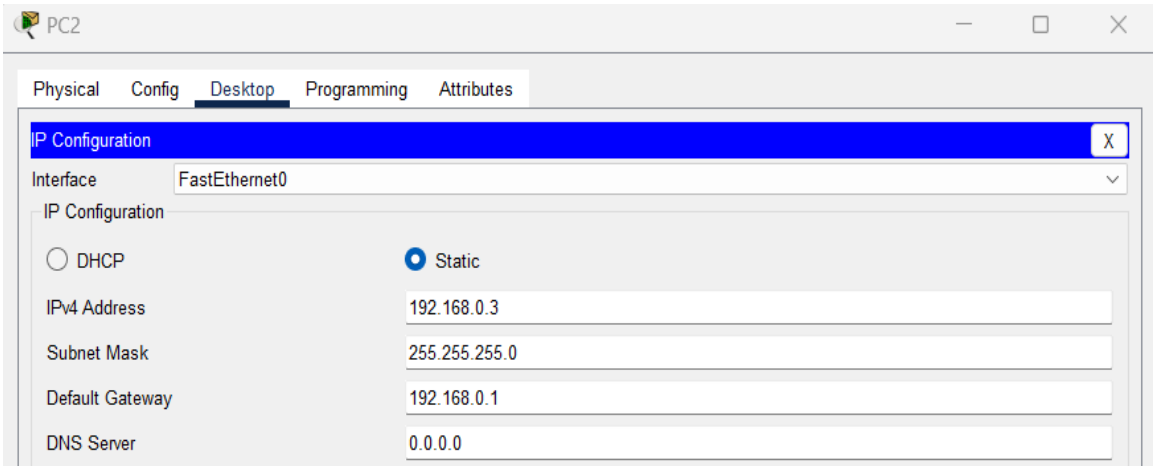
4. On Routers 3 and 4, configure static routes to reach the network connected to Router 1 using the appropriate next-hop IP addresses.

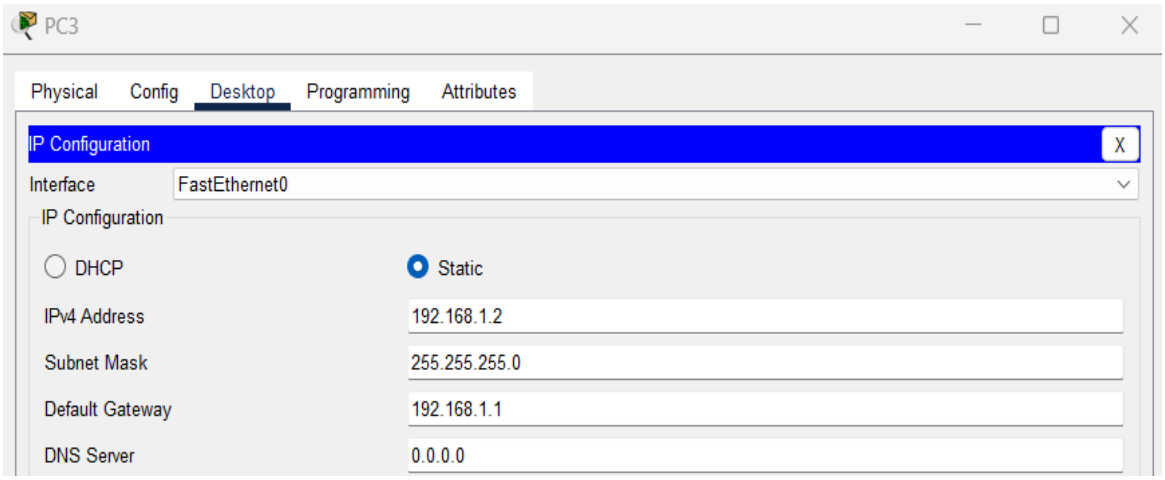


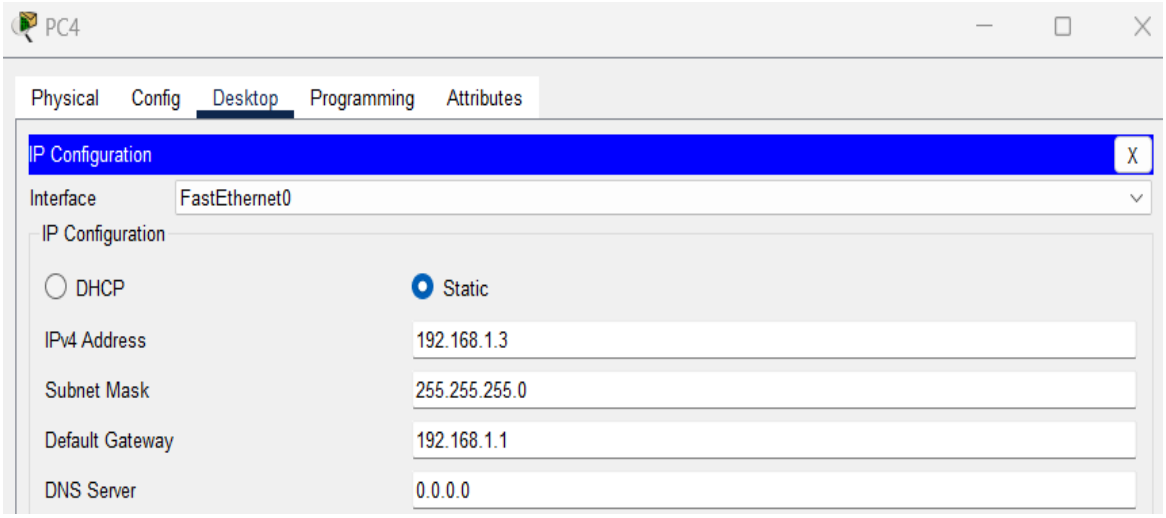


5. On each PC, configure the default gateway setting to point to the IP address of the router interface it is directly connected to.

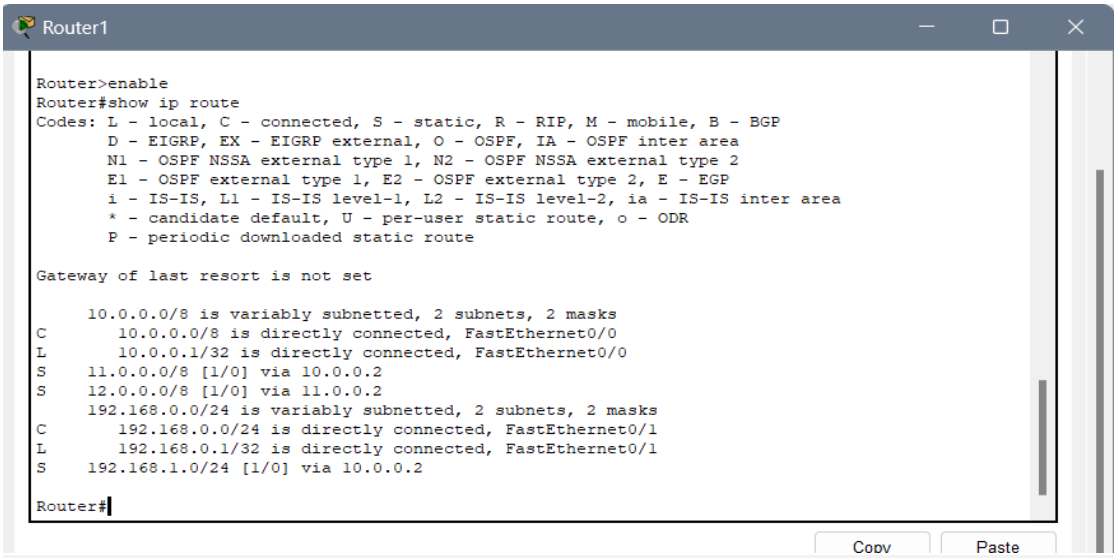


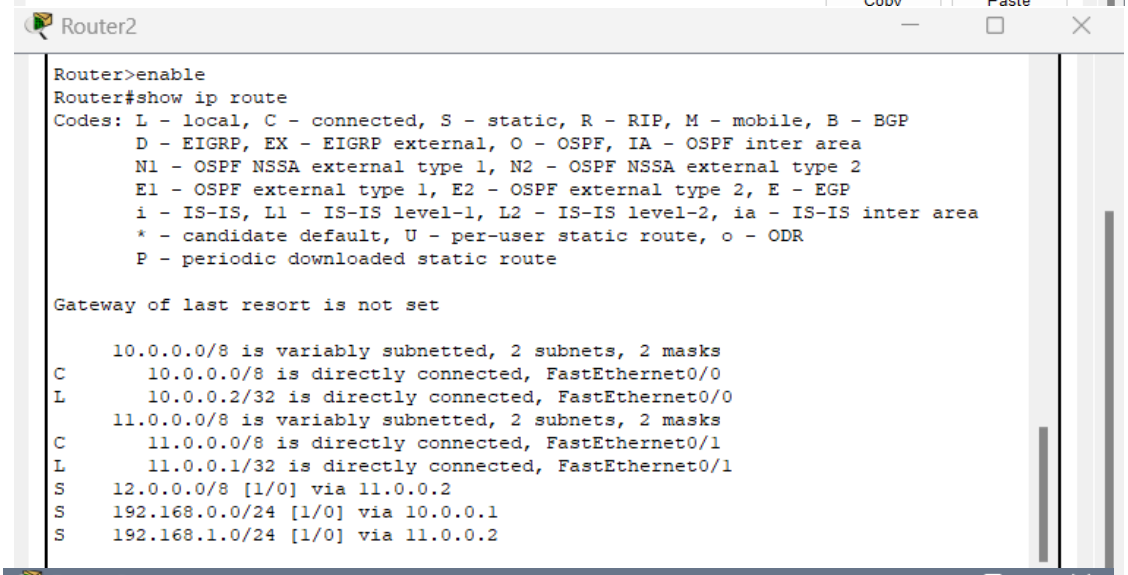

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.0.1
DNS Server
0.0.0.0

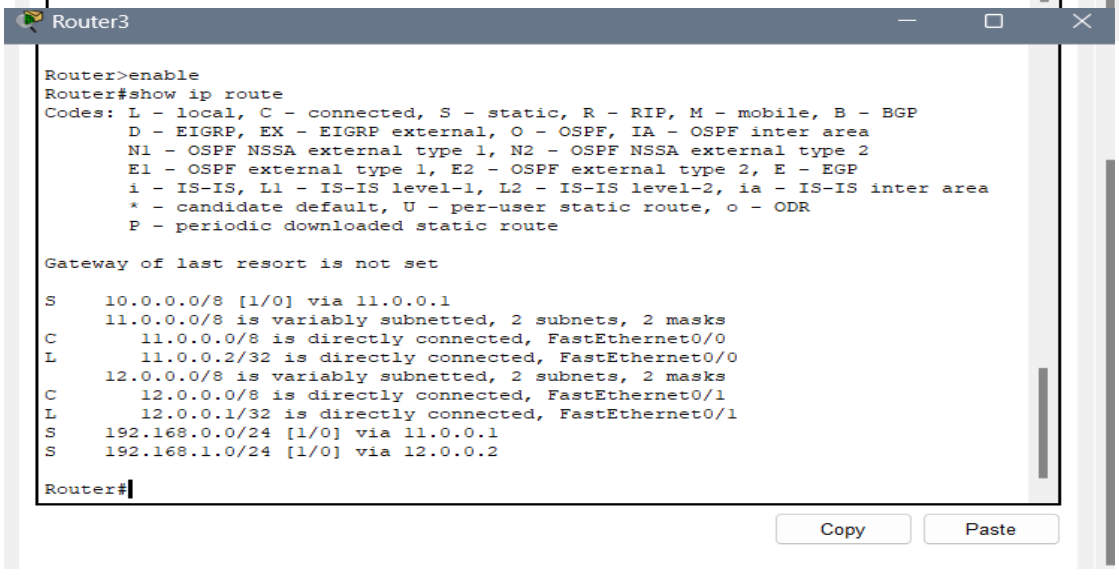

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0

6. Verify the static routing configuration by checking the routing tables on each router using the "show Ip route" command.







Router4

IOS Command Line Interface

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 11.0.0.1
S     11.0.0.0/8 [1/0] via 12.0.0.1
      12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        12.0.0.0/8 is directly connected, FastEthernet0/0
L        12.0.0.2/32 is directly connected, FastEthernet0/0
S     192.168.0.0/24 [1/0] via 12.0.0.1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, FastEthernet0/1
L        192.168.1.1/32 is directly connected, FastEthernet0/1

Router#
```

7. Test end-to-end connectivity between the PCs by attempting to ping from one to another.

PC1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=124
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124
Reply from 192.168.1.2: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

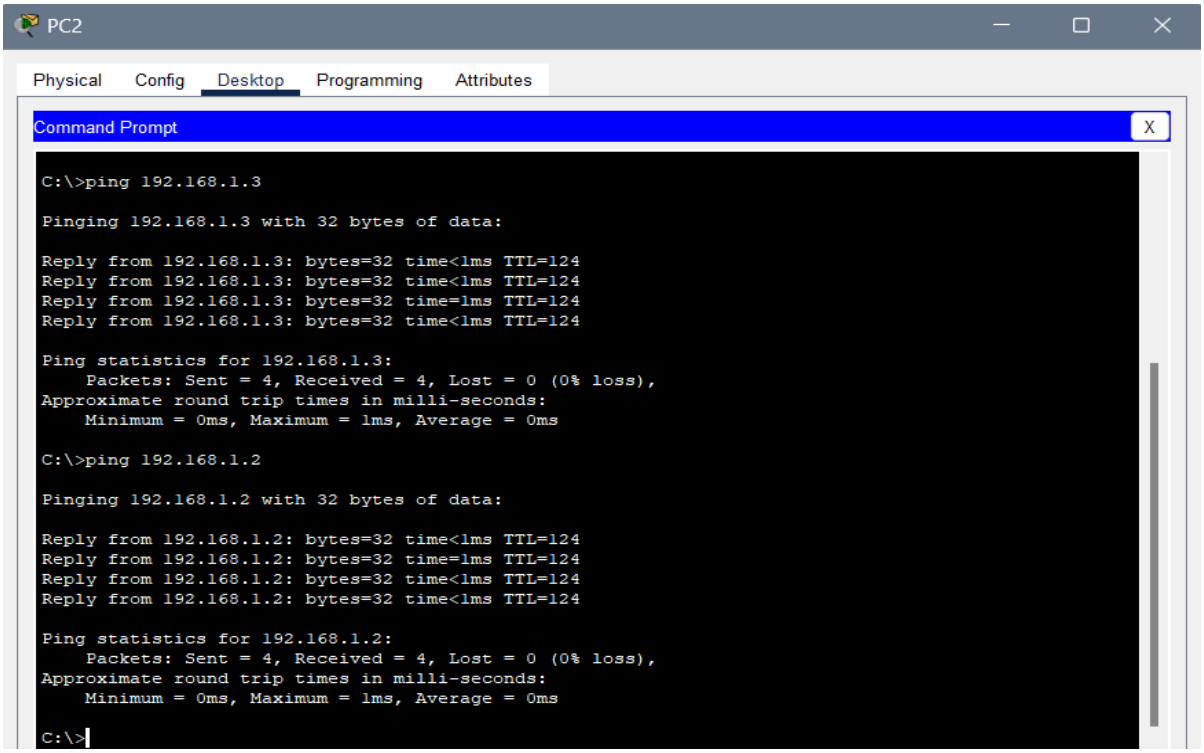
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=124
Reply from 192.168.1.3: bytes=32 time<1ms TTL=124
Reply from 192.168.1.3: bytes=32 time=1ms TTL=124
Reply from 192.168.1.3: bytes=32 time=10ms TTL=124

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```



5. Conclusion

This practical work implements a network with static routing using Cisco Packet Tracer. By configuring static routes, we enabled communication between devices in different router. The manual configuration of static routes provided insight into how smaller networks can be managed effectively without the need for dynamic routing protocols.

The successful ping results confirmed that the static routes were configured correctly, and all devices were able to connect. Doing this practical work shows the importance of careful accurate IP addressing in network.

6. References

- YouTube video:

  Static Routing Overview: https://www.youtube.com/watch?v=DNG7QLyCiEc&t=1s

  static routing with Connecting 4 routers with explanation: https://www.youtube.com/watch?v=rZw_b0wpQ00&list=PLB57s6OrG8LjS4rXfvYZd95H5oHPabDcF&index=4